

Part-1:- DevBoard Manual Deployment (Without Docker Compose)

This guide documents the manual deployment of the DevBoard application on an AWS EC2 instance using Docker containers only. Docker Compose is intentionally not used.

1. Prerequisites

- Launch Ubuntu EC2 instance.
- Install Docker.
- Clone the repository:
- `git clone https://github.com/LondheShubham153/devboard`
- `cd devboard`

2. Clean Up

- `docker stop $(docker ps -aq)`
- `docker rm $(docker ps -aq)`
- `docker network prune -f`
- `docker volume prune -f`

3. Create Docker Network and Volume

- `docker network create devboard_default`
- `docker volume create devboard_pgdata`

4. Create .env

- `POSTGRES_USER=postgres`
- `POSTGRES_PASSWORD=postgres123`
- `POSTGRES_DB=devboard`
- `POSTGRES_PORT=5432`
- `PORT=8080`
- `POSTGRES_URL=postgres://postgres:postgres123@postgres:5432/devboard?sslmode=disable`

5. Run PostgreSQL

- `docker run -d --name postgres --network devboard_default -e POSTGRES_USER=postgres -e POSTGRES_PASSWORD=postgres123 -e POSTGRES_DB=devboard -p 5432:5432 -v devboard_pgdata:/var/lib/postgresql/data -v $(pwd)/init/postgres:/docker-entrypoint-initdb.d postgres:17`

6. Backend Dockerfile (Multi-stage)

- FROM golang:1.24 AS builder
- WORKDIR /app
- COPY go.mod go.sum ./
- RUN go mod download
- COPY . .
- RUN CGO_ENABLED=0 GOOS=linux go build -o devboard-backend .
- FROM alpine:latest
- WORKDIR /app
- COPY --from=builder /app/devboard-backend .
- EXPOSE 8080
- CMD ["/devboard-backend"]

7. Build Backend

- cd backend
- docker build -t devboard-backend:v1 .

8. Run Backend

- docker run -d --name backend --network devboard_default --env-file ../env -p 8081:8080 devboard-backend:v1

9. Frontend Dockerfile (Multi-stage)

- FROM node:22 AS builder
- WORKDIR /app
- COPY package*.json ./
- RUN npm install --legacy-peer-deps
- COPY . .
- RUN npm run build
- FROM node:22-alpine
- WORKDIR /app
- COPY package*.json ./
- RUN npm install --legacy-peer-deps
- RUN npm install -g vite
- COPY --from=builder /app/dist ./dist
- COPY vite.config.js ./
- EXPOSE 5173
- CMD ["npm","run","preview","--","--host","0.0.0.0","--port","5173"]

10. Build Frontend

- cd frontend
- docker build -t devboard-frontend:v1 .

11. Run Frontend

- `docker run -d --name frontend --network devboard_default -p 5173:5173 devboard-frontend:v1`

12. Verify

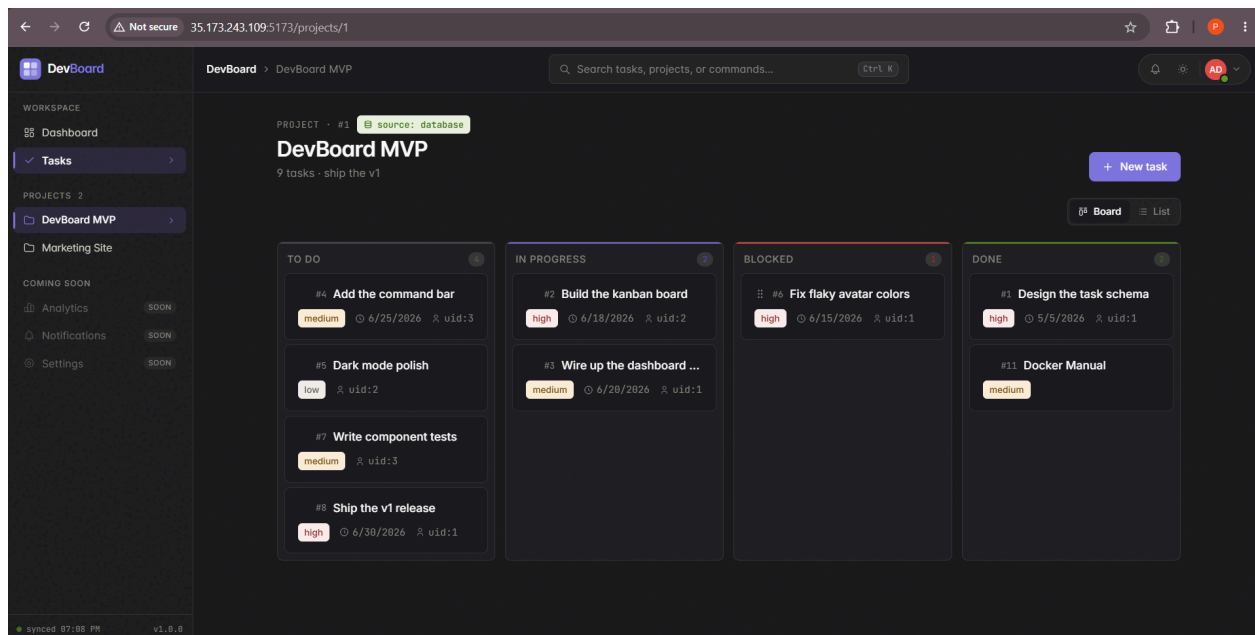
- `docker ps`
- `curl http://localhost:8081/health`
- `curl http://localhost:8081/projects`
- `curl "http://localhost:8081/tasks?project_id=1"`
- Open: `http://<EC2-PUBLIC-IP>:5173`

13. Check Database

- `docker exec -it postgres psql -U postgres -d devboard`
- `\dt`
- `SELECT * FROM projects;`
- `SELECT * FROM tasks;`
- `SELECT COUNT(*) FROM tasks;`
- `\q`

14. Restart After EC2 Reboot

- `docker start postgres`
- `docker start backend`
- `docker start frontend`



id updated_at	title	description	project_id	assignee_id	status	priority	due_date	created_at	
1 22 12:15:37.102053+00	Design the task schema	projects, tasks, statuses	1	1	done	high	2026-05-05	2026-07-22 12:15:37.102053+00	2026-07-
2 22 12:15:37.102053+00	Build the kanban board	drag and drop columns	1	2	in_progress	high	2026-06-18	2026-07-22 12:15:37.102053+00	2026-07-
3 22 12:15:37.102053+00	Wire up the dashboard hero	velocity + chip stats	1	1	in_progress	medium	2026-06-20	2026-07-22 12:15:37.102053+00	2026-07-
4 22 12:15:37.102053+00	Add the command bar	global search	1	3	todo	medium	2026-06-25	2026-07-22 12:15:37.102053+00	2026-07-
5 22 12:15:37.102053+00	Dark mode polish	grain + gradients	1	2	todo	low		2026-07-22 12:15:37.102053+00	2026-07-
6 22 12:15:37.102053+00	Fix flaky avatar colors	hash collision on initials	1	1	blocked	high	2026-06-15	2026-07-22 12:15:37.102053+00	2026-07-
7 22 12:15:37.102053+00	Write component tests	Vitest + testing-library	1	3	todo	medium		2026-07-22 12:15:37.102053+00	2026-07-
8 22 12:15:37.102053+00	Ship the v1 release	tag and announce	1	1	todo	high	2026-06-30	2026-07-22 12:15:37.102053+00	2026-07-
9 22 13:39:22.981823+00	Docker Manual	Docker Manual	1		done	medium		2026-07-22 13:39:17.713155+00	2026-07-
10 22 13:39:42.109576+00	Draft the launch blog post		2	2	todo	medium	2026-06-22	2026-07-22 12:15:37.102053+00	2026-07-
11 22 13:39:43.302189+00	Hero illustration		2	3	todo	low		2026-07-22 12:15:37.102053+00	2026-07-
(11 rows)									
(END)									

i-0778679ba26f928e2 (Web-server)
PublicIPs: 35.173.243.109 PrivateIPs: 172.31.87.57

Part-2: DevBoard Deployment Guide (With Docker Compose)

This guide documents the deployment of the DevBoard application on AWS EC2 using Docker Compose.

1. Prerequisites

- Launch Ubuntu EC2 instance.
- Install Docker.
- Install Docker Compose Plugin.
- git clone https://github.com/LondheShubham153/devboard
- cd devboard

2. Clean Up

- docker stop frontend backend postgres
- docker rm frontend backend postgres
- docker image prune -f (Optional)

3. Create .env

- POSTGRES_USER=postgres
- POSTGRES_PASSWORD=postgres123
- POSTGRES_DB=devboard
- POSTGRES_PORT=5432
- PORT=8080
-

POSTGRES_URL=postgres://postgres:postgres123@postgres:5432/devboard?sslmode=disable

4. Backend Dockerfile

- Use backend multi-stage Dockerfile.

5. Frontend Dockerfile

- Use frontend multi-stage Dockerfile.

6. Create docker-compose.yml

- Create services: postgres, backend, frontend.
- Create volume: devboard_pgdata.
- Create network: devboard-network.

7. Build and Start

- `docker compose up --build -d`

8. Verify

- `docker ps`
- `docker logs backend`
- `docker logs frontend`
- `docker logs postgres`

9. Test APIs

- `curl http://localhost:8081/health`
- `curl http://localhost:8081/projects`
- `curl "http://localhost:8081/tasks?project_id=1"`

10. Access Application

- `http://<EC2-PUBLIC-IP>:5173`

11. Check Database

- `docker exec -it postgres psql -U postgres -d devboard`
- `\dt`
- `SELECT * FROM projects;`
- `SELECT * FROM tasks;`
- `SELECT COUNT(*) FROM tasks;`
- `\q`

12. Useful Commands

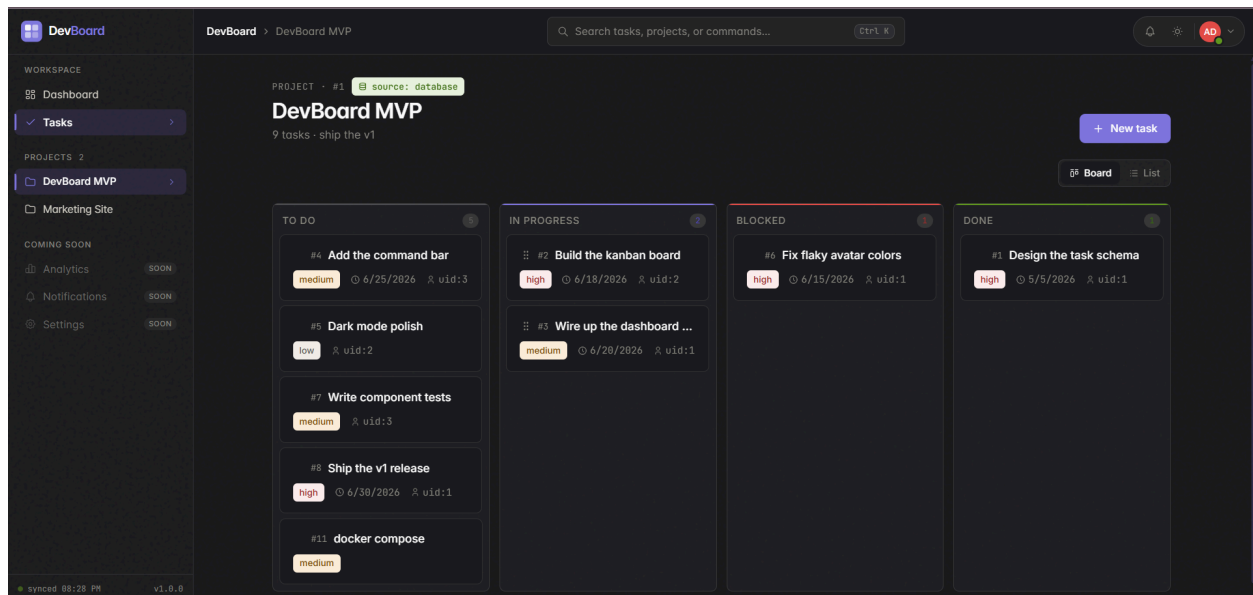
- `docker compose build`
- `docker compose up -d`
- `docker compose ps`

- docker compose logs -f
- docker compose stop
- docker compose restart
- docker compose down
- docker compose down -v

13. Restart After EC2 Reboot

- docker compose start
- docker compose up -d

Docker compose



```

local devboard devboard_pgdata
local devboard_pgdata
ubuntu@ip-172-31-87-57:~/devboard$ docker exec -it postgres psql -U postgres -d devboard
psql (17.10 (Debian 17.10-1.pgd13+1))
Type "help" for help.

devboard=# SELECT COUNT(*) FROM tasks;
 count
-----
      11
(1 row)

devboard=# SELECT id, title, status
FROM tasks
ORDER BY id;
 id | title | status
----+-----+-----
  1 | Design the task schema | done
  2 | Build the kanban board | in progress
  3 | Wire up the dashboard hero | in progress
  4 | Add the command bar | todo
  5 | Dark mode polish | todo
  6 | Fix flaky avatar colors | blocked
  7 | Write component tests | todo
  8 | Ship the v1 release | todo
  9 | Draft the launch blog post | in progress
 10 | Hero illustration | todo
 11 | docker compose | todo
(11 rows)

devboard=#

```